

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.02, beta.var.scale = 0.8\n')
```

```
## sdev = 0.02, beta.var.scale = 0.8
```

Key Summary Statistics

```
cat(paste0('```\n', '

--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.4689  0.5056  0.5139  0.5134  0.5217  0.5506

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09556 0.14111 0.13688 0.17278 0.87944
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09833 0.14500 0.14030 0.17667 0.75778
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.102139 0.031111 1.000000
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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0005556 0.0055556 0.0083333 0.0774919 0.0238889 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0061111 0.0150000 0.162549 0.118333 1.000000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.2428  0.2661  0.2744  0.2739  0.2817  0.3056

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.05278 0.10278 0.10560 0.14222 0.89278
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.05556 0.10667 0.10855 0.14667 0.71611
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.001667 0.053889 0.412369 1.000000 1.000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0050000 0.2468052 0.3816667 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.0000  0.1161  1.0000  0.6953  1.0000  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.07944 0.10722 0.11500 0.11577 0.12278 0.17889

```

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beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.03014 0.07889 0.08444 0.12000 0.89111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.03222 0.08333 0.08798 0.12556 0.58222
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.7322 0.8939  0.7503  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.000000 0.001667 0.241536 0.343194 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.2417 1.0000  0.7165  1.0000  1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.07611 0.09833 0.10778 0.11070 0.12000 0.19889

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01833 0.05611 0.06764 0.09722 0.83278
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1         1
  Modality 5 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02000	0.05944	0.07069	0.10333	0.55444

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3500	0.7506	0.6412	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2618	0.5897	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.8717	1.0000	0.8252	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.04000	0.05556	0.06333	0.06747	0.07611	0.14111

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.041111	0.055889	0.084444	0.818333

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01000	0.04444	0.05931	0.09111	0.56944

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.1511	0.5733	0.5463	0.9872	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7228	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
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    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2548  0.5758  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.8206  1.0000  0.8275  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.03889 0.06278 0.07611 0.07843 0.09167 0.20278

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.025556 0.044439 0.067222 0.787778
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1

gamma.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.004444 0.028056 0.047834 0.073333 0.527222
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.2622  0.3902  0.7867  1.0000
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.6494 1.0000  1.0000  0.9995  1.0000  1.0000
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.1906  0.2293  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.4656  0.8144  0.6740  0.9839  1.0000

--- Summary for K = 7 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.2444 0.2761  0.2850  0.2882  0.2961  0.4461

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

0.000000 0.002778 0.021667 0.047136 0.058889 0.775556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.003333	0.023333	0.044549	0.062222	0.433889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.03389	0.25092	0.47389	1.00000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.4850	0.5583	0.5470	0.6261	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.00000	0.0313	0.00000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.0200	0.2517	0.4906	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.03111	0.05111	0.06167	0.06478	0.07444	0.15333

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.029444	0.046039	0.069444	0.713889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.005556 0.034444 0.050980 0.079444 0.453333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.000000 0.05667 0.28464 0.59056 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.4772 1.0000 1.0000 0.9929 1.0000 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.1327 0.0000 1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.2983 0.6094 0.5431 0.8172 1.0000

```

--- Summary for K = 9 ---

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Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.3978 0.4294 0.4406 0.4411 0.4522 0.5294

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.003889 0.025556 0.051440 0.060556 0.729444
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1 1 1 1 1 1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1 1 1 1 1 1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1 1 1 1 1 1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1 1 1 1 1 1

```

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.004444 0.028889 0.050347 0.067778 0.436111
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.0000 0.0000 0.1006 0.1928 1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.4033 0.4650 0.4974 0.5467 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.00000 0.00000 0.00000 0.01524 0.00000 0.50444
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.

```

```
0.00000 0.00000 0.00000 0.08178 0.15500 0.73833
', '\n`~`'))
```

```
## ‘‘‘
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.4689  0.5056  0.5139  0.5134  0.5217  0.5506
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.00000 0.09556 0.14111 0.13688 0.17278 0.87944
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.00000 0.09833 0.14500 0.14030 0.17667 0.75778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.000000 0.005000 0.008333 0.102139 0.031111 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0005556 0.0055556 0.0083333 0.0774919 0.0238889 1.0000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.000000 0.006111 0.015000 0.162549 0.118333 1.000000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.2428  0.2661  0.2744  0.2739  0.2817  0.3056
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.00000 0.05278 0.10278 0.10560 0.14222 0.89278
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.05556 0.10667 0.10855 0.14667 0.71611
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.001667 0.053889 0.412369 1.000000 1.000000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000000 0.0005556 0.0050000 0.2468052 0.3816667 1.0000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.1161 1.0000 0.6953 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.07944 0.10722 0.11500 0.11577 0.12278 0.17889
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.03014 0.07889 0.08444 0.12000 0.89111
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03222 0.08333 0.08798 0.12556 0.58222
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.7322  0.8939  0.7503  1.0000  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.001667 0.241536 0.343194 1.000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.2417  1.0000  0.7165  1.0000  1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.07611 0.09833 0.10778 0.11070 0.12000 0.19889
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01833 0.05611 0.06764 0.09722 0.83278
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02000 0.05944 0.07069 0.10333 0.55444
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.3500  0.7506  0.6412  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2618  0.5897  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.8717  1.0000  0.8252  1.0000  1.0000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.04000 0.05556 0.06333 0.06747 0.07611 0.14111
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008889 0.041111 0.055889 0.084444 0.818333
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01000 0.04444 0.05931 0.09111 0.56944
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1511 0.5733 0.5463 0.9872 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.7228 1.0000 1.0000 1.0000 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.2548 0.5758 1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.8206 1.0000 0.8275 1.0000 1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.03889 0.06278 0.07611 0.07843 0.09167 0.20278
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.025556 0.044439 0.067222 0.787778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.028056 0.047834 0.073333 0.527222
##   Modality 2 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.2622  0.3902  0.7867  1.0000
##   Modality 3 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6494  1.0000  1.0000  0.9995  1.0000  1.0000
##   Modality 4 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1906  0.2293  1.0000
##   Modality 5 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.4656  0.8144  0.6740  0.9839  1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2444  0.2761  0.2850  0.2882  0.2961  0.4461
##
## beta.ar:
##   Modality 1 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.002778 0.021667 0.047136 0.058889 0.775556
##   Modality 2 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.003333 0.023333 0.044549 0.062222 0.433889
##   Modality 2 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.03389 0.25092 0.47389 1.00000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.4850 0.5583 0.5470 0.6261 1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.0313 0.0000 1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0200 0.2517 0.4906 1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.03111 0.05111 0.06167 0.06478 0.07444 0.15333
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.029444 0.046039 0.069444 0.713889
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.034444 0.050980 0.079444 0.453333
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.05667 0.28464 0.59056 1.00000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4772 1.0000 1.0000 0.9929 1.0000 1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.1327 0.0000 1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.2983 0.6094 0.5431 0.8172 1.0000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3978 0.4294 0.4406 0.4411 0.4522 0.5294
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.003889 0.025556 0.051440 0.060556 0.729444
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.028889 0.050347 0.067778 0.436111
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.1006 0.1928 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000 0.4033 0.4650 0.4974 0.5467 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.01524 0.00000 0.50444
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.08178 0.15500 0.73833
##
## ‘‘‘

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.